AN EXPLICIT MINIMAL THREE-UNIVERSAL CLASS OF COUNTABLE LINEAR ORDERS

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Editorial note. All linear orders are countable. The class is interpreted up to isomorphism, and J means a proper isomorphism-invariant subclass as specified by the universality definition.

Editorial note. The construction chooses representatives by endpoint behavior and the finite-block decomposition; it does not assert an isomorphism classification of every input linear order.

Editorial note. 2026-10-11 材料收口：复用既有 H2、来源许可和近邻判断；核对原命题与作者证明起止、所指上下文及装配范围齐备。未重证作者结论；沿用既有非重复依据。

Editorial scope. One original exact result. Native construction and load-bearing local dependencies preserved; named classical Scott-analysis inputs explicit.

Allowed standard prerequisites (editorial).

- Countable linear orders and infinitary back-and-forth. The supplied interval criterion for asymmetric back-and-forth; the unique countable dense endpoint-free order Q; successor relation is Pi1 and fixed finite successor configurations admit their usual infinitary descriptions.
- Scott-sentence preservation. A Pi_alpha Scott sentence is preserved in the indicated alpha-back-and-forth comparison. The short target-specific minimality lemma is included.

In this article we give a comprehensive Scott analysis of the class of linear orderings using three definitions of rank. Given a structure $\mathcal{A}$ the *(parameterless) Scott rank* $SR(\mathcal{A})$, is the least α, such that the automorphism orbits of all tuples in $\mathcal{A}$ are $\Sigma^{\text{in}}_\alpha$ definable; the *parameterized Scott rank* $pSR(\mathcal{A})$, is the least ordinal α such that there is a parameter $\bar{p} \in A^{<\omega}$ such that $SR((\mathcal{A}, \bar{p})) = \alpha$; and the *Scott sentence complexity* of $\mathcal{A}$, $SSC(\mathcal{A})$ is the least of the complexities Π^{in}_α, $\Sigma^{\text{in}}_\alpha$, $d\text{-}\Sigma^{\text{in}}_\alpha$ such that $\mathcal{A}$ has a Scott sentence of one of these complexities. While the latter is not an

0.1. **Preliminaries.** We assume that the reader is familiar with the basic techniques of Scott analysis and with linear orderings in general. For background we refer to Montalbán's upcoming book [4] and the standard reference by Ash and

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Knight [1]. Let us highlight the following lemma which we will use as basic tools in our proofs without reference.

Lemma 0.1 ([1, Lemma 15.8])**.** *Suppose $\mathcal{A}, \mathcal{B} \in \mathrm{LO}$. Then $\mathcal{A} \leq_1 \mathcal{B}$ if and only if $\mathcal{A}$ is infinite or at least as large as $\mathcal{B}$. For $\alpha > 1$, $\mathcal{A} \leq_\alpha \mathcal{B}$ if and only if for every $1 \leq \beta < \alpha$ and every partition of $\mathcal{B}$ into intervals $\mathcal{B}_0, \ldots, \mathcal{B}_n$, with endpoints in $\mathcal{B}$, there is a partition of $\mathcal{A}$ into intervals $\mathcal{A}_1, \ldots, \mathcal{A}_n$ with endpoints in $\mathcal{A}$, such that $\mathcal{B}_i \leq_\beta \mathcal{A}_i$.*

Notice that this lemma particularly implies that $(\mathcal{A}, \bar{a}) \leq_\alpha (\mathcal{B}, \bar{b})$ if and only if $(-\infty, a_0) \leq_\alpha (-\infty, b_0)$, $(a_{|\bar{a}-1|}, \infty) \leq_\alpha (b_{|\bar{b}-1|}, \infty)$ and for all $0 \leq i < |\bar{a} - 1|$ $(a_i, a_{i+1}) \leq_\alpha (b_i, b_{i+1})$ [1, Lemma 15.7].

Definition 0.2. Let $K_0 \subseteq K_1$ be isomorphism invariant classes of structures. The class K_0 is *α-universal* for K_1 if for every $\mathcal{A} \in K_1$ there is $\mathcal{B} \in K_0$ such that $\mathcal{A} \leq_\alpha \mathcal{B}$.

Theorem 0.3. *The following class is 3-universal for the class of linear orderings $K = \{\omega + k, k + \omega^*, k + \mathbb{Z} + k' : k, k' \in \mathbb{N}\} \cup \{\omega + \omega^*\} \cup \{(\sum_{i \in k} n_i + m_i \cdot \mathbb{Q}) + n_{k+1} : n_i, m_i \in \mathbb{N}, n_i > m_{i-1}, m_{i+1}\} \cup \{k : k \in \mathbb{N}\}$. There is no 3-universal class $J \subset K$.*

Lemma 0.4. *If K_0 is α-universal for K_1, then K_0 contains all structures in K_1 with Π^{in}_α Scott sentences.*

证明. Suppose otherwise and consider $\mathcal{A} \in K_1 \setminus K_0$ with a Π^{in}_α Scott sentence. By α-universality, there is $\mathcal{B} \in K_0$ with $\mathcal{A} \leq_\alpha \mathcal{B}$. But then $\mathcal{A} \cong \mathcal{B}$ and thus $\mathcal{A} \in K_0$, a contradiction. □

证明. Given a linear order $\mathcal{L}$ we can write $\mathcal{L} = W + \mathcal{K} + R$ with W well ordered or empty, R reverse well ordered or empty, and $\mathcal{K}$ empty or without a greatest or least element.

The element in K that $\mathcal{L}$ is $\leq_3$-below will depend on the sizes of W, and R and the order type of $\mathcal{K}$. We will deal with these cases independently.

Case 1: W is infinite and R is finite. In this case, $W = \omega + \alpha$ for some ordinal α, so $\mathcal{L} = \omega + \mathcal{K}' + k$ for some $k \in \mathbb{N}$ and $\mathcal{K}'$ is either empty or without greatest element. We claim that $\mathcal{L} \leq_3 \omega + k$. It is sufficient to show that $\omega + \mathcal{K}' \leq_3 \omega$. The $\exists$-player always wins the corresponding back-and-forth game by playing the following strategy. In the first round they play isomorphically on the initial ω. Because of that they only need to win the back-and-forth game for $\omega \leq_2 \omega + \mathcal{K}'$. Say the $\forall$-player picks elements such that $\omega + \mathcal{K}'$ is partitioned into n intervals $\mathcal{A}_i$, each of cardinality m_i for $i < n$. The $\exists$-player must pick a partition of ω such that for all intervals $\mathcal{B}_i$, each of cardinality k_i, $\mathcal{A}_i \leq_1 \mathcal{B}_i$. This inequality holds if and only if $m_i \geq k_i$. Given this inequality, the intervals formed by picking the first n elements of ω clearly form the desired partition.

Case 2: W is finite and R is infinite. This case is analogous to the first case.

Case 3: Both W and R are infinite. In this case, we can write $\mathcal{L} = \omega + \mathcal{K}' + \omega^*$. We claim that $\mathcal{L} \leq_3 \omega + \omega^*$. The winning strategy for the $\exists$-player is similar to the one for case 1. In the first play they play isomorphically to reduce the game to showing that $\omega + \omega^* \leq_2 \omega + \mathcal{K}' + \omega^*$. They can then win this game by picking partitions with smaller intervals, exactly as above.

Case 3: Both W and R are finite. There are two subcases with different behavior here. In both we assume that $\mathcal{K}$ has no greatest and least element, as if it was empty, then $\mathcal{L}$ would be finite and thus isomorphic to a structure in K.

Subcase 1: There are arbitrarily large successor chains in $\mathcal{K}$. Here we claim that $\mathcal{K} \leq_3 \mathbb{Z}$, which gives that $\mathcal{L} \leq_3 k + \mathbb{Z} + k'$ for $k, k' \in \mathbb{N}$. The $\exists$-player always wins the corresponding game by playing the following strategy. Assume the $\forall$-player plays an ordered tuple with the distance between least and greatest element n. Then the $\exists$-player plays a successor chain of size n in $\mathcal{K}$ such that $\mathcal{K} = \mathcal{K}_1 + n + \mathcal{K}_2$ where both $\mathcal{K}_1$ and $\mathcal{K}_2$ are infinite. It is then sufficient to show that $\omega^* \leq_2 \mathcal{K}_1$ and $\omega \leq_2 \mathcal{K}_2$. In these games, similar to the cases above, the $\exists$-player just needs to ensure that they play partitions of smaller size then the $\forall$-player and this is possible by the same strategy.

Subcase 2: There is an $n \in \mathbb{N}$ such that no element in $\mathcal{K}$ has n successors. In this case, we consider the block relation $\sim$ given by $a \sim b$ if and only if $[a, b]$ and $[b, a]$ are finite. By assumption every block is a linear order of size n or less. As no block can be infinite, no two blocks can be successors in the quotient $\mathcal{K}/\sim$. Furthermore, as there is no least or greatest element, there are no least or greatest blocks in $\mathcal{K}/\sim$. Therefore, $\mathcal{K}/\sim = \mathbb{Q}$. In total, this means that $\mathcal{K} = \sum_{q \in \mathbb{Q}} i_n(q)$ where $i_n : \mathbb{Q} \to \{1, \cdots, n\}$. If there are only finitely many blocks of size n , we may write $\mathcal{K}$ as a sum $\mathcal{K}_1 + n + \cdots + n + \mathcal{K}_l$ where no block of size n occurs in any of the $\mathcal{K}_i$. Repeating this process as necessary, we can write $\mathcal{K}$ as $\mathcal{K}_1 + n_1 + \mathcal{K}_2 + n_2 + \ldots \mathcal{K}_l$ where each $\mathcal{K}_i$ is given by $\mathcal{K}_i = \sum_{q \in (b_i, t_i)} i_n(q)$ such that $\{q \in (b_i, t_i) : i_n(q) = m_i\}$ is infinite for $m_i = \max \text{range}(i_n \restriction (b_i, t_i))$. Furthermore, we have that $m_i > n_i$ and if $i > 1$, $m_i > n_{i-1}$.

We now show that $\mathcal{L} = k + \mathcal{K}_1 + n_1 + \ldots \mathcal{K}_l + k' \leq_3 k + m_1 \cdot \mathbb{Q} + n_1 \ldots m_l \cdot \mathbb{Q} + k'$. It is enough to show that $\mathcal{K}_i = \sum_{q \in (b_i, t_i)} i_n(q) \leq_3 m_i \cdot \mathbb{Q}$. The $\exists$-player always wins this game by playing the following strategy. Every element the $\forall$-player plays in the first play is in a block of size m_i. The $\exists$-player can respond by playing elements in the same position in blocks of size m_i. The resulting intervals in $\mathcal{K}_i$ are of the form $l_1 + \sum_{q \in (b_i^j, t_i^j)} i_n(q) + l_2$ where $b_i^j > b_i$ and $t_i^j < t_i$. The corresponding intervals in $m_i \cdot \mathbb{Q}$ are of the form $l_1 + m_i \cdot \mathbb{Q} + l_2$. Winning the game thus comes down to finding a winning strategy for the game $m_i \cdot \mathbb{Q} \leq_2 \sum_{q \in (b_i^j, t_i^j)} i_n(q)$. Given a partition in $\sum_{q \in (b_i^j, t_i^j)} i_n(q)$, the $\exists$-player needs to find a partition in $m_i \cdot \mathbb{Q}$ such

that all intervals are smaller than the intervals in the partition picked by the $\forall$-player. They can do that using the fact that the finite blocks are densely ordered and always of maximum size.

This shows that K is 3-universal. Standard arguments show that each element of K has a Π^{in}_3 Scott sentence. Therefore, by Lemma 0.4, there is no $J \subset K$ such that J is 3-universal. □

References

[1] C. Ash and J. Knight. *Computable Structures and the Hyperarithmetical Hierarchy*. Studies in Logic and the Foundations of Mathematics144,2000.

[2] R. Alvir and D. Rossegger. The complexity of Scott sentences of scattered linear orders. *Journal of Symbolic Logic*85(3)(2020),1079–1101.

[3] C.J. Ash. Recursive labelling systems and stability of recursive structures in hyperarithmetical degrees. *Transactions of the American Mathematical Society*298(2)(1986),497–514.

[4] A. Montalban. *Computable Structure Theory: Beyond the Arithmetic*. Draft,2021. `https://math.berkeley.edu/~antonio/CSTpart2_DRAFT.pdf`.

[5] David Gonzalez and Antonio Montalb'an. *The ω-Vaught's Conjecture*. Native bibliography entry supplied in flattened.bbl.

[6] Antonio Montalb'an. Classes of structures with no intermediate isomorphism problems. *Journal of Symbolic Logic*81 (2016),127–150.